

JUNKAI HUANG

☎ (615) 275-8369 | ✉ junkaih@andrew.cmu.edu | 🏠 www.junkai.site | 🔗 linkedin.com/in/junkaih

Objective: With my expertise in machine learning and 3D computer vision, I aim to develop and deploy innovative ML solutions that solve real-world industrial challenges and create impactful products.

EDUCATION

Carnegie Mellon University, Robotics Institute	Pittsburgh, PA
• MS in Robotics GPA: 4.17 / 4.33	August 2023 - May 2025 (Expected)
• Selected Coursework: Advanced Computer Vision, Learning-based Image Synthesis, Physics-based Rendering	
The Hong Kong University of Science and Technology (HKUST)	Hong Kong SAR
• BSc in Computer Science and Mathematics GPA: 3.96 / 4.3	September 2019 - August 2023
• Selected Coursework: Deep Learning in Computer Vision, Machine Learning, Advanced Computer Graphics, Algorithms, OOP and Data Structures, Operating Systems, Linear Algebra, Probability	

WORK EXPERIENCE

Rivian Automotive, Inc	Palo Alto, CA
<i>Software Engineer Intern (ML) - Perception & Pose</i>	May 2024 - August 2024
- Built and improved the data auto-labeling pipelines, enabling accurate labeling of Rivian's autonomous driving dataset. - Developed the first 3D occupancy prediction model taking camera and LiDAR inputs. The model was trained on labeled Rivian data achieving mIoU 49.2 on test set, used to label the rest 80% unlabeled Rivian data. - Improved 3D point cloud ground labeling accuracy on inclined and uneven surfaces by 21% using Patchwork++. - Built 3D point cloud visualization tools in MLflow, reducing issue detection time by 30%.	
Sebit Company Limited	Hong Kong SAR
<i>AI/ML Developer Intern</i>	June 2022 - August 2022
- Built and maintained the data annotation, model training and evaluation pipeline for faster iteration. The dangerous object detection models trained with my pipeline was deployed across 12 escalators at two major malls. - Automated dataset creation, model configuration, training, and evaluation with Jenkins. Each stage has clear running logs and breakpoint recovery, increasing the model version iteration speed by 85%. - Integrated pre-trained YOLOv4 into manual data annotation workflow to improve the annotation speed by 50%.	

RESEARCH EXPERIENCE

Echoes of the Coliseum: Towards 3D Live streaming of Human-Centric Events First author	<i>Under Review</i>
<i>3D reconstruction, Gaussian Splatting, 3D Pose Estimation, Distributed System, VR/AR</i> - Proposed a novel framework processing multi-view images for immersive novel-view live streaming. - Proposed to optimize 3D Gaussians leveraging human priors, distributed processing, coarse-to-fine approach, etc., achieving photo-realistic 3D reconstruction of live events within 5s per frame. - Project leader: leading a cross-functional 7 people team, contributing to 80% of the code.	
Instance Neural Radiance Field First author 📄 Paper 📺 Video	ICCV 2023
<i>3D Instance Segmentation, Neural Rendering</i> Proposed one of the first 3D instance segmentation pipeline for NeRFs, leveraging 3D proposal-based discrete mask prediction, 3D-2D mask projection, and instance identity matching, achieving mIoU 43.0 (23% higher than baseline).	
NeRF-RPN: A general framework for object detection in NeRFs First author 📄 Paper 📺 Video	CVPR 2023
<i>3D Object Detection, Neural Rendering, CNN, Transformer, RPN</i> - Proposed the first 3D object detection framework for NeRFs by exploiting multi-scale 3D neural volumetric representation, region proposal network (RPN), various backbone architectures, RPN head designs and loss functions. - Built a large-scale public indoor NeRF benchmark dataset that consists of 400+ scenes and 6000+ objects.	
DragGaussian: Point-based 3D Gaussian Manipulation 🌐 Webpage	CMU Spring 2024
<i>3D Gaussian Editing, Diffusion Model</i> Proposed a diffusion model based 3D Gaussian editing method, taking 3D point dragging signals as input, using 2D supervision to manipulate 3D Gaussians, achieving intuitive and direct 3D asset editing.	